

Vanguard / Ethos Open Source  
Research Division

March 23, 2026

Editorial Board  
Annals of Mathematics  
Princeton University

Dear Editors,

I am writing to submit our manuscript, "*The Collatz Attractor: Thermodynamic Destiny in Discrete Dynamical Systems*", for consideration in *Annals of Mathematics*.

This paper reframes the 80-year-old Collatz Conjecture ( $3n+1$  problem) not as a purely arithmetic puzzle, but as a discrete integer model of non-equilibrium thermodynamics. By mapping the operations  $n/2$  and  $3n+1$  to the Master Equation ( $S = H + P$ ) and the dissipation of free energy, we demonstrate that the sequence mathematically behaves as a gradient descent over a friction landscape toward the  $S_{max}$  terminal attractor.

Given the journal's history of publishing groundbreaking reductions of seemingly intractable problems, we believe this topological and thermodynamic translation offers the formal algebraic path required to finally close the conjecture.

Thank you for your consideration.

Sincerely,

Rafael D. De Paz